

ANNAPURNA 360 — JavaScript Source Code

Frontend Interaction Engine Documentation

```
/* =====  
  ANNAPURNA 360  
  Frontend interaction engine  
===== */  
  
/* ===== GLOBAL DATA ===== */  
  
let selectedFoodName = "Rice";  
let selectedFoodIcon = "🍚";  
  
let batches =  
  JSON.parse(  
    localStorage.getItem("annapurna360_batches")  
  ) || [];  
  
/* ===== FOOD SELECTION ===== */  
  
function selectFood(name, icon){  
  
  selectedFoodName = name;  
  selectedFoodIcon = icon;  
  
  document.getElementById("selectedFood").innerHTML =  
    `${icon} ${name} selected`;  
  
  document.getElementById("predictionFood").value =  
    name;  
  
  document.getElementById("passportFood").innerHTML =  
    `${icon} ${name}`;  
  
}  
  
/* ===== ROLE ===== */  
  
function selectRole(role){  
  
  const messages = {  
  
    provider:  
      "👨‍🍳 Food Provider Mode selected. Let's prevent unnecessary cooking and rescue surplus  
food.",  
  
    farmer:  
      "👨‍🌾 Farmer / FPO Mode selected. Let's find the right pathway for your agricultural  
surplus.",  
  
    recipient:
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        "🍲 Recipient Mode selected. You can manage your food requirement and capacity.",

        volunteer:
        "🚲 Volunteer Mode selected. You can support pickup and delivery.",

        admin:
        "🏠 Admin Mode selected. You can monitor district-level food impact."

    };

    alert(messages[role]);

}

/* ===== NAVIGATION ===== */

function goToPrediction(){

    document
        .getElementById("prediction")
        .scrollIntoView({
            behavior:"smooth"
        });

}

function goToDonation(){

    document
        .getElementById("donation")
        .scrollIntoView({
            behavior:"smooth"
        });

}

function goToDashboard(){

    document
        .getElementById("dashboard")
        .scrollIntoView({
            behavior:"smooth"
        });

}

/* ===== AI DEMAND PREDICTION ===== */

function predict(){

    const people =
        Number(
            document.getElementById("people").value
        );

    const previous =
        Number(

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        document.getElementById("previousMeals").value
    );

    if(people <= 0){

        document.getElementById("result").innerHTML =
            "⚠️ Please enter a valid number of people.";

        return;
    }

    /*
    Simple explainable MVP logic.

    We add a small buffer instead of
    repeating historical overproduction.
    */

    const recommended =
        Math.ceil(
            (people * 1.05) / 10
        ) * 10;

    const lower =
        Math.max(
            people,
            recommended - 10
        );

    const avoided =
        Math.max(
            0,
            previous - recommended
        );

    document.getElementById("result").innerHTML = `

    <h3>🤖 AI Recommendation</h3>

    <div style="
        font-size:35px;
        font-weight:bold;
        color:#e65100;
        margin:10px 0;
    ">
        ${lower} - ${recommended} meals
    </div>

    <p>
        👥 Expected people:
        <strong>${people}</strong>
    </p>

    <p>
        📦 Previous preparation:

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        <strong>${previous}</strong>
    </p>

    <p>
        🔄 Potential overproduction avoided:
        <strong>${avoided} meals</strong>
    </p>

    <p>
        💡 The recommendation uses expected demand
        and historical preparation as an explainable
        starting point.
    </p>

    `;
}

/* ===== SURPLUS DETECTION ===== */

function detectSurplus(){

    const prepared =
        Number(
            document.getElementById("prepared").value
        );

    const consumed =
        Number(
            document.getElementById("consumed").value
        );

    const storage =
        document.getElementById("storage").value;

    const preparedTime =
        document.getElementById("preparedTime").value;

    const surplus =
        Math.max(
            0,
            prepared - consumed
        );

    let status;

    if(storage === "Unknown"){

        status =
            "🔴 Human safety review required before redistribution.";

    }
    else{

        status =
            "🟡 Potentially suitable for further human assessment.";

    }
}

```



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document.getElementById("passportQuantity").innerText =
    quantity + " portions";

document.getElementById("passportFood").innerText =
    selectedFoodIcon + " " + selectedFoodName;
}

/* ===== SAVE BATCH ===== */

function saveBatch(){

    const batch = {

        id:
            document.getElementById("batchId").innerText,

        food:
            selectedFoodName,

        quantity:
            document.getElementById(
                "passportQuantity"
            ).innerText,

        created:
            new Date().toLocaleString()

    };

    batches.push(batch);

    localStorage.setItem(
        "annapurna360_batches",
        JSON.stringify(batches)
    );

    alert(
        "✅ Food rescue batch saved successfully!"
    );
}

/* ===== FARMER ANALYSIS ===== */

function analyseFarm(){

    const harvest =
        Number(
            document.getElementById("harvest").value
        );

    const market =

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        Number(
            document.getElementById("market").value
        );

const produce =
    document.getElementById("produce").value;

const surplus =
    Math.max(
        0,
        harvest - market
    );

document.getElementById("farmerResult").innerHTML = `

    <h3>🌾 ${produce}</h3>

    <div style="
        font-size:30px;
        font-weight:bold;
        color:#2e7d32;
    ">
        ${surplus} kg potential surplus
    </div>

    <br>

    <p>
        🌱 Recommended pathway order:
    </p>

    <p>
        🛒 FPO / Market Channel
        →
        🏢 Authorized Processing
        →
        🍌 Appropriate Food Channel
        →
        ♻️ Permitted Recovery
    </p>

    <small>
        The correct pathway depends on the produce,
        condition, regulations and authorized partners.
    </small>

`;
}

/* ===== LANGUAGE ===== */

function changeLanguage(){

    const language =
        document.getElementById("language").value;

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```
const heroText =
    document.getElementById("heroText");

if(language === "telugu"){

    heroText.innerText =
        "AI ఆధారిత ఆహార రక్షణ మరియు ఆహార వ్యర్థాల నివారణ వేదిక";

}

else if(language === "hindi"){

    heroText.innerText =
        "AI आधारित खाद्य बचाव और खाद्य अपशिष्ट रोकथाम प्लेटफॉर्म";

}

else if(language === "tamil"){

    heroText.innerText =
        "AI அடிப்படையிலான உணவு மீட்பு மற்றும் உணவு கழிவு தடுப்பு தளம்";

}

else if(language === "kannada"){

    heroText.innerText =
        "AI ಆಧಾರಿತ ಆಹಾರ ರಕ್ಷಣೆ ಮತ್ತು ಆಹಾರ ತ್ಯಾಜ್ಯ ತಡೆ ವೇದಿಕೆ";

}

else{

    heroText.innerText =
        "AI Powered Food Rescue & Waste Prevention Platform";

}

}

/* ===== VOICE ===== */

function readPage(){

    if(
        !("speechSynthesis" in window)
    ){

        alert(
            "Voice reading is not supported in this browser."
        );

        return;

    }

    speechSynthesis.cancel();
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const text =
    document.body.innerText
    .replace(/\s+/g, " ")
    .substring(0, 5000);

const speech =
    new SpeechSynthesisUtterance(text);

speech.rate = 0.9;

speech.pitch = 1;

speech.volume = 1;

speechSynthesis.speak(speech);
}

/* ===== AUTO DEMO ===== */

function demoMode(){

    document.getElementById("people").value = 500;

    document.getElementById("previousMeals").value = 610;

    document.getElementById("prepared").value = 650;

    document.getElementById("consumed").value = 520;

    goToPrediction();

    setTimeout(
        predict,
        500
    );

}

/* ===== INITIALIZATION ===== */

document.addEventListener(
    "DOMContentLoaded",
    function(){

        selectFood(
            "Rice",
            "🍚"
        );

    }
);

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        Give the page a small entrance effect.  
    */  
  
    document.body.classList.add(  
        "loaded"  
    );  
  
    }  
);
```